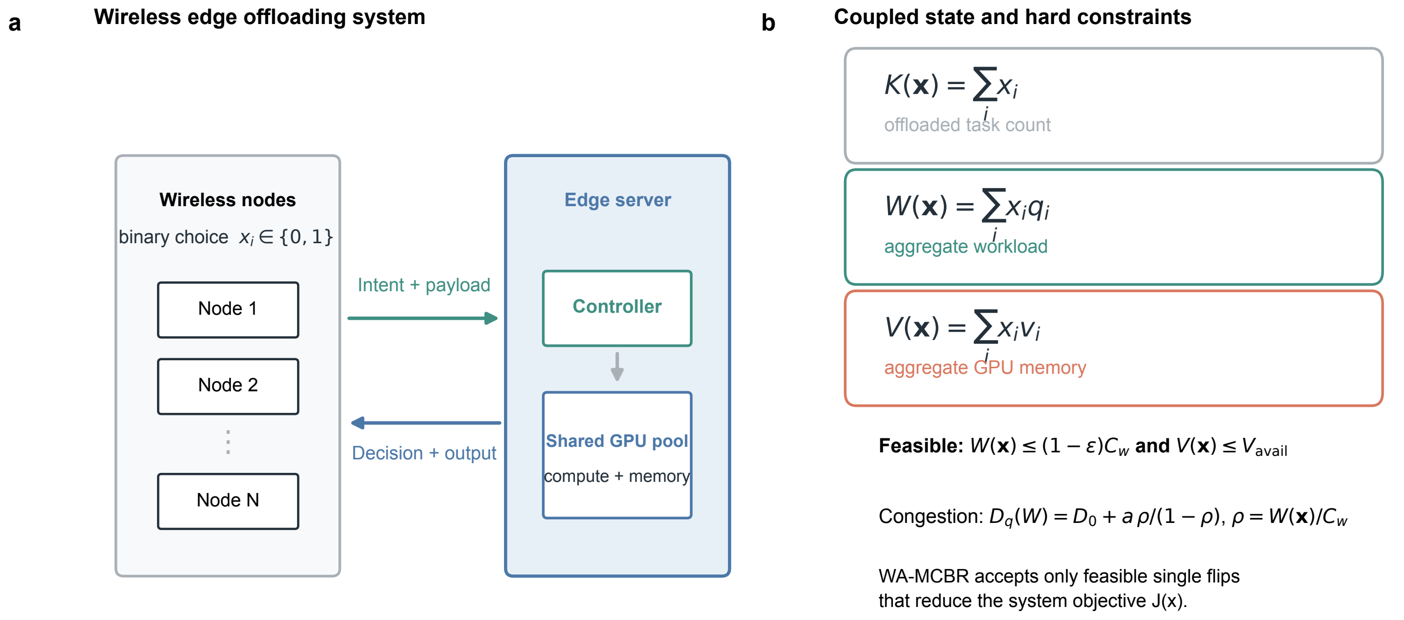
a
Wireless edge offloading system
Wireless nodes
binary choice $x_i \in \{0,1\}$
Node 1
Node 2
Node N
Intent + payload
Decision + output
Edge server
Controller
Shared GPU pool
compute + memory
b
Coupled state and hard constraints
$K(\mathbf{x}) = \sum_i x_i$
offloaded task count
$W(\mathbf{x}) = \sum_i x_i q_i$
aggregate workload
$V(\mathbf{x}) = \sum_i x_i v_i$
aggregate GPU memory
Feasible: $W(\mathbf{x}) \leq (1-\varepsilon)C_w$ **and** $V(\mathbf{x}) \leq V_{\mathrm{avail}}$
Congestion: $D_q(W) = D_0 + a\,\rho/(1-\rho)$, $\rho = W(\mathbf{x})/C_w$
WA-MCBR accepts only feasible single flips
that reduce the system objective J(x).